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Sustainable CO₂ Capture from Gas Flaring Using Calcium Hydroxide

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Abstract

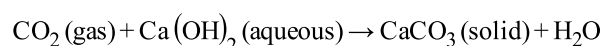
Gas flaring is a widely used practice in the oil and gas industry, primarily employed to safely dispose of excess hydrocarbons that cannot be processed, stored, or transported. While this method is essential for operational safety and pressure regulation, it comes with a significant environmental cost—namely, the emission of large volumes of carbon dioxide (CO₂), a potent greenhouse gas. CO₂ emissions from gas flaring contribute substantially to global warming and climate change, making it imperative to develop and implement sustainable mitigation strategies.

This paper introduces a novel and environmentally sustainable approach to CO₂ capture from gas flaring operations using calcium hydroxide (Ca (OH)₂) as a chemical scrubbing agent. The proposed system is designed to not only ensure the safe combustion of excess gases but also to actively capture and neutralize the CO₂ produced during the flaring process. By integrating a CO₂ scrubbing mechanism into the flaring system, this method significantly reduces the environmental footprint of gas flaring and aligns with global efforts to combat climate change.

The system architecture comprises three main components: a Ground Flare Assembly, a CO₂ Scrubbing Unit, and a set of automated control mechanisms. The process begins with the delivery of fuel gas to the burner through a dedicated gas line. The burner, equipped with a temperature sensor, ignites the gas and continuously monitors the combustion temperature. If the sensor detects optimal combustion conditions (i.e., high temperature), it triggers the closure of a shutter and the opening of an On/Off valve to maintain gas flow. Conversely, if the temperature drops indicating incomplete or failed combustion the shutter remains open, and the valve closes to safely vent unburned gas, preventing hazardous buildup.

The Ground Flare Assembly is engineered for controlled and enclosed combustion. It includes multiple burners, a flow sensor to regulate gas input, and a fresh air blower that supplies the necessary oxygen to sustain efficient combustion. The exhaust gases generated comprising CO₂ and other by-products are channeled through a main pipeline to the scrubbing unit.

The CO₂ Scrubbing Unit is the core of the emission reduction system. It utilizes a solution of calcium hydroxide (Ca (OH)₂) to chemically react with CO₂ in the flue gases. The gases pass through a series of three scrubbers, where they are exposed to the Ca (OH)₂ solution. The primary chemical reaction involved is:



This reaction converts gaseous CO_2 into calcium carbonate (CaCO_3), a stable and non-toxic solid that can be safely stored or repurposed in various industries, such as construction or agriculture. The scrubbers are equipped with level sensors to monitor the volume of the $\text{Ca}(\text{OH})_2$ solution, ensuring that the system operates at peak efficiency. Once the solution becomes saturated with CaCO_3 , it is discharged through an outlet valve, and the treated exhaust gases now significantly reduced in CO_2 content are vented through an exhaust valve.

The integration of calcium hydroxide scrubbing into the gas flaring process offers multiple environmental, operational, and economic benefits. Environmentally, it directly reduces the amount of CO_2 released into the atmosphere, contributing to climate change mitigation. The transformation of CO_2 into a solid form also provides a method of long-term sequestration, effectively removing it from the atmospheric cycle. Operationally, the system enhances safety by ensuring complete combustion and preventing the release of unburned gases. Economically, the by-product calcium carbonate can be sold or reused, turning a waste stream into a valuable resource.

Moreover, the system is designed with real-time monitoring and feedback controls, including CO_2 sensors, temperature sensors, and flow regulators. These components work together to maintain optimal operating conditions, adjust chemical dosing, and ensure the highest possible scrubbing efficiency. The use of automation reduces the need for manual intervention and enhances the reliability and consistency of the process.

In conclusion, the proposed CO_2 capture system using calcium hydroxide scrubbing represents a sustainable, efficient, and scalable solution to the environmental challenges posed by gas flaring. It not only addresses the urgent need to reduce greenhouse gas emissions but also aligns with broader sustainability goals and regulatory frameworks. By converting harmful emissions into useful by-products and integrating advanced monitoring technologies, this system exemplifies how innovation in industrial processes can lead to both environmental stewardship and economic opportunity. As the oil and gas industry continues to evolve in response to climate imperatives, such technologies will play a critical role in shaping a cleaner, more sustainable future.

Introduction

Gas flaring is a routine yet critical operation in the oil and gas industry, used to safely burn off excess hydrocarbons that cannot be stored or processed due to operational constraints or safety concerns. While flaring ensures the safe disposal of volatile gases, it also contributes significantly to environmental degradation by releasing large volumes of carbon dioxide (CO_2) into the atmosphere. CO_2 is a major greenhouse gas, and its accumulation in the atmosphere is directly linked to global warming and climate change. As industries worldwide face increasing pressure to reduce their carbon footprint, innovative and sustainable solutions for emission control have become a top priority.

Traditional flaring systems focus primarily on combustion efficiency and safety, often overlooking the environmental consequences of CO_2 emissions. In response to this challenge, the integration of CO_2 capture technologies into flaring systems presents a promising opportunity to mitigate the environmental impact of these operations. Among the various methods available, chemical scrubbing using calcium hydroxide ($\text{Ca}(\text{OH})_2$) has emerged as a practical and cost-effective solution for capturing CO_2 from flue gases.

This manuscript introduces a comprehensive system designed to capture CO_2 emissions from gas flaring using $\text{Ca}(\text{OH})_2$ as a scrubbing agent. The system combines advanced combustion monitoring with a multi-stage scrubbing process to ensure both operational safety and environmental sustainability. Key components include a ground flare assembly for controlled combustion, a fresh air blower for oxygen supply, and a scrubbing unit where the chemical reaction between CO_2 and $\text{Ca}(\text{OH})_2$ takes place, forming calcium carbonate (CaCO_3) and water.

The proposed system is equipped with sensors and control mechanisms that regulate temperature, gas flow, and chemical dosing in real time. These features ensure optimal combustion and scrubbing efficiency while minimizing waste and operational costs. Additionally, the by-product of the scrubbing process—solid CaCO_3 —can be safely stored or repurposed in industries such as construction and agriculture, aligning with circular economy principles.

By integrating CO_2 capture into the gas flaring process, this system offers a sustainable pathway for reducing greenhouse gas emissions in the oil and gas sector. It not only addresses environmental concerns but also enhances operational efficiency and opens new avenues for resource recovery and cost savings.

Design Methodology

The design methodology for capturing carbon dioxide (CO_2) from gas flaring using calcium hydroxide ($\text{Ca}(\text{OH})_2$) is a multi-stage, integrated system that combines combustion control, gas flow regulation, chemical scrubbing, and real-time monitoring. This methodology is developed to ensure maximum efficiency in CO_2 capture while maintaining operational safety and environmental sustainability. The system is composed of several key components, each playing a vital role in the overall process, from the initial combustion of excess hydrocarbons to the final neutralization and removal of CO_2 . As per the [fig-1](#).

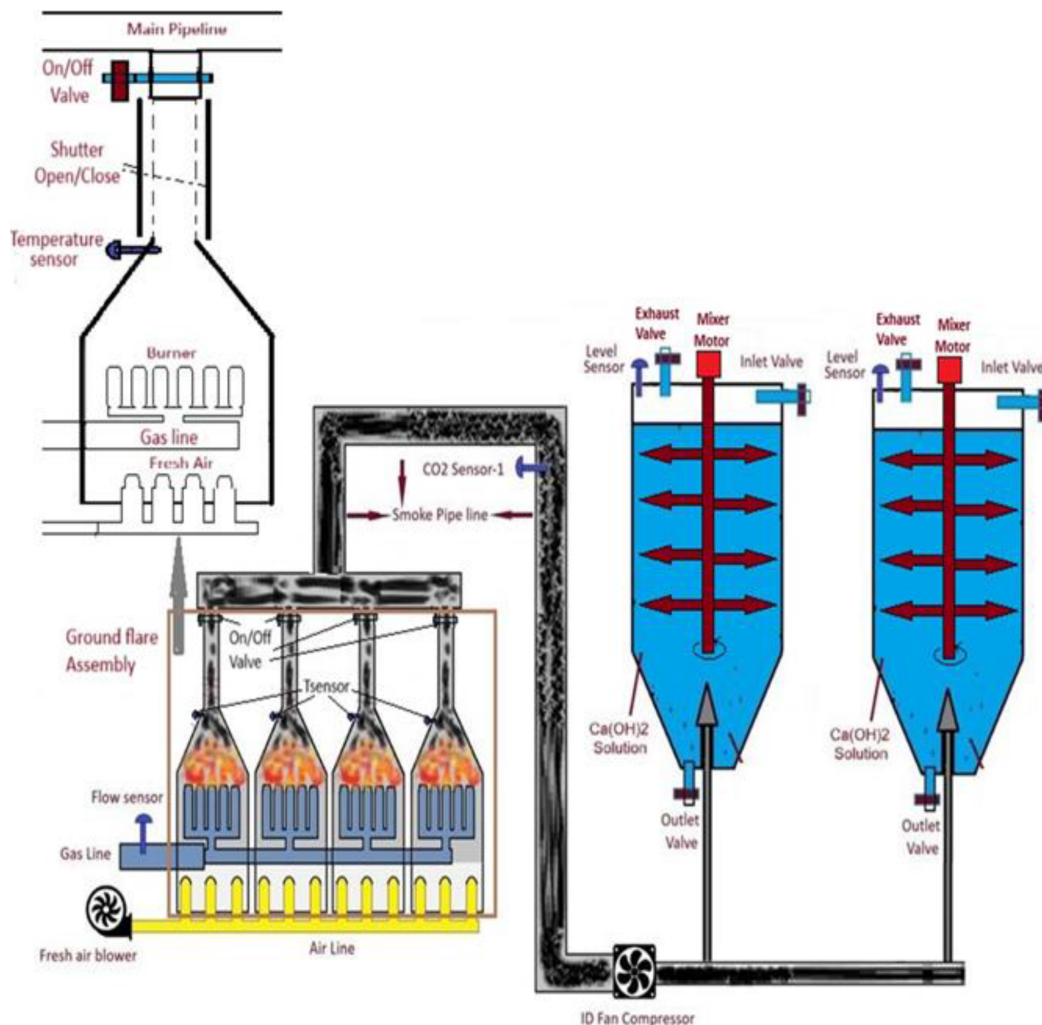


Figure 1

Ground Flare Assembly and Combustion Control

The process begins with the Ground Flare Assembly, which is designed to safely combust excess or waste hydrocarbons. This assembly includes multiple burners enclosed within a fennel assembly, which creates a controlled combustion environment. The burners are connected to a flue gas line that channels untreated gases from the process line to the flare system.

A temperature sensor is installed at the top of the fennel assembly to monitor the combustion temperature. This sensor is critical for regulating the shutter and On/Off valve mechanisms. When the sensor detects a high temperature indicating proper combustion, the shutter closes to containing the combustion gases, and the On/Off valve opens to allow the flow of burned gases into the passage line. If the temperature drops, signaling incomplete combustion, the shutter remains open to vent unburned gases safely into the atmosphere, and the valve closes to prevent their entry into the scrubbing system.

The materials used in the Ground Flare Assembly are selected for durability and resistance to high temperatures and corrosive gases. Stainless steel, Inconel alloys, and ceramic coatings are commonly used to ensure long-term reliability and safety.

Fresh Air Blower and Oxygen Supply

Efficient combustion requires a consistent supply of oxygen. The Fresh Air Blower is responsible for drawing air from the atmosphere and directing it to the base of the Ground Flare Assembly. This airflow supports the ignition and complete combustion of the fuel gas, reducing the risk of releasing unburned hydrocarbons.

The blower operates continuously during the flaring process and is constructed from corrosion-resistant materials such as stainless steel or aluminum alloys. Its design ensures that the air is evenly distributed to all burners, maintaining combustion efficiency and system safety.

Passage Line and Gas Transport

Once combustion occurs, the resulting exhaust gases—containing CO₂ and other by-products—are transported through a Passage Line to the scrubbing unit. This conduit is designed to maintain the necessary pressure and flow rate for efficient gas movement.

The passage line is typically made from carbon steel, stainless steel, or HDPE, depending on the temperature and chemical composition of the gases. It connects the Ground Flare Assembly to the Induced Draft (ID) Fan, which facilitates the movement of gases into the scrubbing system.

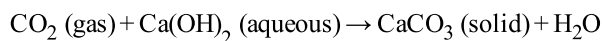
Induced Draft Fan and Flow Regulation

The ID Fan plays a crucial role in maintaining the flow of exhaust gases through the system. It creates a vacuum that pulls the gases from the flare assembly into the scrubbing unit. The fan operates at a nominal speed of 750 RPM, but its speed can be adjusted based on the gas flow rate, which is monitored by a Flow Sensor.

The flow sensor ensures that the ID Fan operates efficiently by providing real-time data on the volume and velocity of the gases. This feedback allows the system to adjust fan speed dynamically, maintaining optimal pressure and flow for effective CO₂ scrubbing.

CO₂ Scrubbing Unit and Chemical Neutralization

The heart of the CO₂ capture system is the Scrubbing Unit, which contains a Ca (OH)₂ Solution Tank. This tank holds a mixture of calcium hydroxide and water, which reacts with CO₂ in the exhaust gases to form calcium carbonate (CaCO₃) and water. The chemical reaction is:



The tank is equipped with Blowing Nozzles that inject exhaust gases into the solution. These nozzles ensure even distribution of gases within the tank, maximizing the surface area for the chemical reaction. Additionally, mixer installed inside the tank to increase the solubility of $\text{Ca}(\text{OH})_2$ in water, allowing for more interaction between CO_2 and $\text{Ca}(\text{OH})_2$.

The tank also includes Inlet and Outlet Valves for managing the solution flow. The Inlet Valve allows fresh $\text{Ca}(\text{OH})_2$ solution to enter the tank, while the Outlet Valve discharges the spent solution and CaCO_3 slurry. A Level Sensor monitors the volume of the solution, ensuring that the tank maintains the correct concentration for effective scrubbing.

Materials used in the tank construction include polypropylene, fiberglass, and stainless steel, with titanium or ceramic coatings applied in high-stress areas to resist corrosion and wear.

CO₂ Sensors and Monitoring

To ensure the effectiveness of the scrubbing process, the system employs two CO₂ Sensors:

- Sensor 1 is in the main passage line before the gases enter the scrubbing tank. It measures the initial concentration of CO₂ in the exhaust gases.
- Sensor 2 is positioned at the top of the $\text{Ca}(\text{OH})_2$ solution tank and measures the CO₂ concentration after scrubbing.

By comparing the readings from both sensors, operators can assess the efficiency of the CO₂ capture process. These sensors use NDIR (Non-Dispersive Infrared) or electrochemical technology and are housed in corrosion-resistant materials such as stainless steel or ceramics.

Post-Reaction Handling and Safety Mechanisms

After the chemical reaction, the resulting CaCO_3 and excess water are removed from the system through the Outlet Valve. This discharge process prevents the accumulation of waste material and allows for the safe disposal or repurposing of calcium carbonate.

A Vent Valve is installed at the top of the tank to release any excess pressure that may build up during the reaction. This safety feature ensures that the system operates within safe pressure limits and prevents potential hazards.

Automation and Control Integration

The entire system is designed for automated operation, with real-time monitoring and feedback loops controlling key parameters such as temperature, gas flow, CO₂ concentration, and solution levels. The integration of sensors and control mechanisms allows for dynamic adjustments to maintain optimal performance.

For example, if Sensor 1 detects a high CO₂ concentration, the system can increase the flow of $\text{Ca}(\text{OH})_2$ solution or adjust the ID Fan speed to enhance scrubbing efficiency. Similarly, if the Level Sensor indicates a low solution volume, the Inlet Valve opens automatically to replenish the tank.

This automation reduces the need for manual intervention, enhances safety, and ensures consistent operation under varying conditions.

Results of the Concept

The combustion of hydrocarbon-rich flare gas is a common practice in oil and gas operations, but it releases significant amounts of carbon dioxide (CO₂), a greenhouse gas. To mitigate environmental impact, one effective method is to capture CO₂ using a lime slurry system, where calcium hydroxide ($\text{Ca}(\text{OH})_2$) reacts with CO₂ to form calcium carbonate (CaCO_3).

For basis calculation and detail concept outlines considered that the chemical basis, material requirements, and tank design for treating CO₂ emissions from the combustion of 1 ton of flare gas.

Flare Gas Composition and CO₂ Production

The flare gas in question consists of methane (CH₄, 75%), ethane (C₂H₆, 10%), propene (C₃H₆, 5%), and hydrogen sulfide (H₂S, 10%). Upon combustion, CH₄, C₂H₆, and C₃H₆ produce CO₂, while H₂S does not contribute to CO₂ formation. Based on stoichiometric calculations:

- CH₄ produces 1 mole of CO₂ per mole,
- C₂H₆ produces 2 moles of CO₂ per mole,
- C₃H₆ produces 3 moles of CO₂ per mole.

From 1 ton (1,000,000 grams) of flare gas, the total CO₂ produced is approximately **2.51 tons (2,507,420.94 grams)**.

Ca (OH)₂ Requirement and Slurry Preparation

To capture this CO₂, calcium hydroxide is used in a 1:1 molar ratio. The required amount of Ca (OH)₂ is approximately **4.22 tons (4,221,195.57 grams)**. For practical application, Ca (OH)₂ is suspended in water to form a slurry. In this case, a concentration of **150 g/L** is chosen, which is suitable for industrial use while maintaining manageable viscosity and mixing properties.

To dissolve or suspend 4.22 tons of Ca (OH)₂ at 150 g/L, a total slurry volume of **28,141.30 liters (28.14 m³)** is required.

Tank Design for Lime Slurry

A cylindrical tank is designed to hold the slurry. Using a height-to-diameter ratio of 1.5 (common for efficient mixing), the tank dimensions are:

- **Diameter:** 2.88 meters
- **Height:** 4.32 meters

This configuration ensures optimal space utilization and mixing dynamics. The tank should be constructed from corrosion-resistant materials such as FRP or HDPE-lined steel to withstand the alkaline nature of the slurry.

Mixing and Operational Considerations

To maintain the slurry in suspension and ensure effective CO₂ absorption, continuous mixing is essential. The estimated mixing power requirement is **56.28 Watts**, based on a standard of 2 W/m³. Mechanical agitators or recirculating pumps can be used to achieve this.

The above calculation is assumption of 1ton of flue gas burning and 150g/L of limestone used.

Economic Summary

The CO₂ capture system using calcium hydroxide offers notable economic advantages for oil and gas operations. By significantly reducing CO₂ emissions from gas flaring, the system helps operators avoid potential carbon taxes, regulatory penalties, and compliance costs associated with environmental legislation. Additionally, the by-product of the scrubbing process calcium carbonate (CaCO₃) is a valuable material that can be repurposed in industries such as construction, agriculture, and manufacturing, creating a potential revenue stream.

The system's design emphasizes durability and low maintenance, using corrosion-resistant materials like stainless steel and Inconel alloys. Its automation and real-time monitoring reduce labor costs and operational downtime. Overall, the integration of this technology not only supports environmental sustainability but also enhances cost-efficiency and long-term profitability for industrial facilities.

Discussion and Conclusion

The integration of calcium hydroxide ($\text{Ca}(\text{OH})_2$) scrubbing into gas flaring operations represents a transformative approach to reducing carbon dioxide (CO_2) emissions in the oil and gas industry. This concept addresses a long-standing environmental challenge flaring of excess hydrocarbons—by not only ensuring safe combustion but also capturing and neutralizing the resulting CO_2 emissions. The system's design, which combines combustion control, chemical scrubbing, and real-time monitoring, has demonstrated both technical feasibility and environmental effectiveness.

From a technical standpoint, the system's modular architecture allows for precise control over each stage of the process. The Ground Flare Assembly ensures complete combustion through temperature-regulated shutters and valves, while the Fresh Air Blower supplies the necessary oxygen to support efficient burning. The Induced Draft (ID) Fan and flow sensors maintain consistent gas movement, ensuring that exhaust gases reach the scrubbing unit under optimal conditions. This level of control minimizes the risk of incomplete combustion and unburned gas release, enhancing both safety and performance.

The core of the system the $\text{Ca}(\text{OH})_2$ scrubbing unit has proven highly effective in capturing CO_2 . The chemical reaction between CO_2 and calcium hydroxide is simple yet powerful, converting gaseous CO_2 into solid calcium carbonate (CaCO_3). This reaction not only removes CO_2 from the exhaust stream but also produces a stable by-product that can be repurposed in various industries. The use of mixer in the tank and blowing nozzles within the scrubber tank increases solubility of limestone with water, gas-liquid interaction time, further improving scrubbing efficiency.

Real-time monitoring through CO_2 sensors and level sensors ensures that the system operates at peak efficiency. Sensor feedback allows for dynamic adjustments to fan speed, solution concentration, and flow rates, creating a responsive and intelligent system. This automation reduces the need for manual intervention and enhances reliability.

Environmentally, the system contributes significantly to climate change mitigation. By capturing CO_2 before it enters the atmosphere, it reduces the greenhouse gas burden and supports global sustainability goals. The conversion of CO_2 into CaCO_3 also aligns with circular economy principles, turning emissions into usable resources.

Economically, the system offers cost savings through reduced carbon penalties and potential revenue from CaCO_3 by-products. Its durable construction and low maintenance requirements further enhance its financial viability.

In conclusion, the CO_2 capture system using calcium hydroxide scrubbing presents a sustainable, efficient, and scalable solution for emission control in gas flaring operations. It combines environmental responsibility with operational excellence and economic benefit. As industries move toward greener practices, this technology stands out as a practical and impactful tool for reducing carbon emissions and promoting a low-carbon future.

Key References

Post-Combustion CO_2 Capture with Calcium and Lithium Hydroxide

This study explores the use of $\text{Ca}(\text{OH})_2$ and LiOH as solid sorbents for CO_2 capture from flue gases. It highlights the effectiveness of $\text{Ca}(\text{OH})_2$ at low temperatures (20–150 °C) and the enhancement of CO_2 uptake in the presence of moisture. <https://www.nature.com/articles/s41598-022-14235-5.pdf>

Higher CO₂ Absorption Using Calcium Hydroxide Nanoparticles

This research introduces nano-sized Ca(OH)₂ particles (10 nm) that significantly increase CO₂ absorption capacity—up to 8 times more than conventional particles. It also discusses the potential for treating acidic gases like SO_x and NO_x. <https://link.springer.com/article/10.1007/s10311-018-0729-4>

Synthetic Calcium Carbonate Production by CO₂ Mineralization

This U.S. Department of Energy project demonstrates how CO₂ captured using Ca(OH)₂ can be converted into high-purity precipitated calcium carbonate (PCC), which has commercial value in industries like construction, paper, and agriculture. https://netl.doe.gov/sites/default/files/netl-file/21CMOG_CU_Wang.pdf

Calcium Carbonate Looping for Carbon Capture

This European research initiative explores calcium looping as a scalable and efficient post-combustion CO₂ capture method, emphasizing the reuse of CaCO₃ in industrial applications. <https://cordis.europa.eu/article/id/206011-calcium-carbonate-looping-carbon-capture-ready-for-precommercial-demonstration>